

Panduan Otomatisasi Pengujian & Rekomendasi Tools AI QA (Istana Laundry ERP)

- > **Dokumen:** AI-Assisted QA Automation & Testing Strategy
- > **Versi:** 1.0 · **Dipublikasikan:** 30 Juli 2026
- > **Target System:** Istana Laundry Management System (Laravel 13, Docker, Sanctum)
- > **Penulis:** Quality Assurance & AI Architecture Team

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1. Visi & Strategi QA Berbasis AI

Seiring dengan pesatnya perkembangan fitur Istana Laundry ERP (POS, Produksi 8-stasiun, Pengadaan, Akuntansi, Payroll, dan Aset), pengujian manual berulang (*regression testing*) berisiko memakan waktu dan melelahkan.

Strategi **AI-Powered QA Testing** bertujuan untuk:

- **Zero-Maintenance Locators:** Menghindari kegagalan test akibat perubahan selector CSS/ID di Blade templates.
- **Natural Language Test Cases:** Menulis pengujian dalam bahasa instruksi manusia (contoh: "*Ketik promo PROMO50 dan pastikan diskon terpotong*").
- **Autonomous Bug Discovery:** AI agent yang secara mandiri menjelajahi alur bisnis, menemukan *edge cases*, dan melaporkan kecacatan UI.
- **Visual AI Inspection:** Memastikan grafik Chart.js, layout Tailwind, dan komponen Blade tampil presisi di layar desktop maupun seluler.

2. Rekomendasi Ecosystem Tools AI QA Testing

2.1 AI E2E Testing & Natural Language Automation (ZeroStep + Playwright)

- **Teknologi: ZeroStep AI** ([@zerostep/playwright](#)) + **Playwright Test**.
- **Keunggulan:** Memungkinkan penulisan pengujian Playwright menggunakan fungsi `ai()`. AI membaca DOM secara dinamis dan mengeksekusi instruksi tanpa memerlukan hardcoded CSS selector (`#btn-submit`).
- **Penggunaan di Istana Laundry:** Pengujian checkout POS, pengajuan refund 4-tahap, dan penginputan payroll karyawan.

2.2 AI Visual Regression Testing (Applitools Eyes / VisualAI)

- **Teknologi:** Applitools Eyes VisualAI / Percy by BrowserStack.
- **Keunggulan:** Menggunakan Computer Vision berbasis AI untuk membandingkan screenshot UI sebelum dan sesudah deployment. Berbeda dari pixel-matching tradisional, VisualAI mengabaikan perbedaan piksel rendering minor dan hanya mendeteksi perubahan visual yang berdampak pada pengguna.
- **Penggunaan di Istana Laundry:** Memeriksa grafik Chart.js pada Executive Dashboard & Laporan Keuangan, serta cetak slip gaji & struk thermal.

2.3 Autonomous Agent Exploratory Testing (Octomind AI)

- **Teknologi:** Octomind.ai / Bugbug.io AI.
- **Keunggulan:** AI Bot yang melakukan crawling otomatis pada aplikasi web, memetakan seluruh rute (/pos, /production, /finance, /hr), dan menghasilkan script Playwright secara otomatis jika menemukan error 404/500 atau form crash.
- **Penggunaan di Istana Laundry:** Pengujian stres pada alur multi-cabang dan permission role.

2.4 Self-Healing Test Locators (Healenium / Playwright AI)

- **Teknologi:** **Healenium** (Plugin Selenium/Playwright) atau **Relicx AI**.
- **Keunggulan:** Apabila struktur HTML Blade berubah (misal class Tailwind berganti), Healenium menganalisis tree DOM dengan algoritma ML dan memperbaiki selector yang patah secara real-time tanpa menggagalkan build CI/CD.

2.5 AI REST API Contract & Fuzzing (Postman Postbot & Schemathesis)

- **Teknologi:** Schemathesis AI + Postman Postbot.
- **Keunggulan:** Membaca spesifikasi OpenAPI/Sanctum REST API ([/api/production](#), [/api/track](#)) dan secara otomatis menghasilkan ratusan skenario request acak (fuzzy testing) untuk menguji ketahanan server terhadap SQL Injection, Null Pointer, dan Unauthorized Bypass.

...

3. Arsitektur Implementasi Script AI QA

```
■ GitHub Actions CI/CD Pipeline ■
```

```
■ 1. Docker Compose Up → PHP 8.4-FPM + Nginx + MySQL 8 + App ■
```

```
■ 2. Playwright + ZeroStep AI ■■■ (Natural Language Prompts) ■■■■■■■■■■ ■
```

```
■ 3. Applitools VisualAI ■■■■■■■■ (UI Visual Comparison) ■■■■■■■■■■■■■■■■■■■■
```

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4. Contoh Playwright + AI Test Script (POS & Billing Flow)

```
import { test, expect } from '@playwright/test';
import { ai } from '@zerostep/playwright';

test.describe('AI Automated POS Checkout & Promo Test', () => {
  test('Cashier can apply manual promo code and process payment', async ({ page }) => {
    // 1. Navigasi ke Halaman Login
    await page.goto('http://localhost:8000/login');

    // 2. Gunakan AI Prompt untuk Login
    await ai('Type "cashier.smd01@istanalaundry.com" into the email field', { page });
    await ai('Type "password" into the password field', { page });
    await ai('Click the login button', { page });

    // 3. Buka Halaman POS
    await page.goto('http://localhost:8000/pos');

    // 4. Instruksi AI Natural Language
    await ai('Select customer "Budi Santoso" from the dropdown or search list', { page });
    await ai('Add item "Cuci Komplit Reguler" with quantity 5 to cart', { page });
    await ai('Enter promo code "PROMO50" in the coupon code box and click Apply', { page });

    // 5. Verifikasi Diskon Terpotong
    const isPromoApplied = await ai('Confirm green success alert "Kupon PROMO50 berhasil diterapkan" is visible', { page });
    expect(isPromoApplied).toBeTruthy();

    // 6. Selesaikan Pembayaran
    await ai('Enter 50000 in the cash payment input', { page });
    await ai('Click the "Proses Bayar & Cetak Nota" button', { page });

    // 7. Verifikasi Struk Muncul
    await ai('Confirm invoice receipt modal or window is displayed', { page });
  });
});
```

5. Contoh Visual AI Test Script (Dashboard & Reports)

```
import { test } from '@playwright/test';
import { Eyes, Target } from '@applitools/eyes-playwright';

test.describe('Visual AI Layout & Chart Verification', () => {
  let eyes: Eyes;

  test.beforeAll(() => {
    eyes = new Eyes();
    eyes.setApiKey(process.env.APPLITOOLS_API_KEY || 'YOUR_API_KEY');
```

```
});

test('Executive Dashboard Charts Visual Check', async ({ page }) => {
  await page.goto('http://localhost:8000/login');
  await page.fill('#email', 'owner@istanalaundry.com');
  await page.fill('#password', 'password');
  await page.click('button[type="submit"]');

  await page.goto('http://localhost:8000/dashboard');

  // Inisialisasi Applitools Eyes Visual Check
  await eyes.open(page, 'Istana Laundry ERP', 'Executive Dashboard Layout');

  // Periksa visual seluruh halaman beserta Chart.js
  await eyes.check('Dashboard Full View', Target.window().fully());

  await eyes.close();
});
});
```

6. Integrasi CI/CD Pipeline (GitHub Actions)

Tambahkan langkah AI QA Testing pada [.github/workflows/ci.yml](#):

```
name: Istana Laundry AI QA Pipeline

on:
  push:
  branches: [ master ]
  pull_request:
  branches: [ master ]

jobs:
  ai-qa-testing:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v2

      - name: Start Environment in Docker
        run: |
          cp .env.example .env
          docker compose up -d --build
          docker compose exec -T app php artisan migrate --seed

      - name: Install Node & Playwright
        run: |
          npm install -g @playwright/test @zerostep/playwright
          npx playwright install --with-deps

      - name: Run AI Natural Language E2E Tests
        env:
          ZEROSTEP_TOKEN: ${ secrets.ZEROSTEP_TOKEN }
        run: npx playwright test tests/e2e/

      - name: Run Schemathesis API Fuzzing
        run: |
          pip install schemathesis
          schemathesis run http://localhost:8000/api/documentation
```

> **Kesimpulan:** Dengan mengadopsi kombinasi **ZeroStep AI**, **Applitools VisualAI**, dan **Schemathesis**, pengujian Istana Laundry ERP menjadi jauh lebih cepat, resilien terhadap perubahan UI Blade, dan menjamin kualitas enterprise secara otomatis di CI/CD.